KHUSHI ADVANI

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Data Scientist with 4+ years of experience translating complex analyses into actionable insights and measurable business impact. Experienced in applying machine learning, predictive analytics, forecasting, and statistical modeling to complex business problems, with hands-on expertise in Python, SQL, advanced ML techniques, time-series modeling, and AI/LLM applications. Proven ability to build data-driven decision systems and partner with senior stakeholders, with experience spanning consulting, healthcare, insurance, and financial services.

EDUCATION

Columbia University — New York City, NY
MS Data Science, GPA: 3.83/4.0 — Expected Dec 2026

Purdue University — West Lafayette, IN
BS Computer Science (Machine Intelligence), GPA: 3.75/4.0 — May 2021

WORK EXPERIENCE

Spinnaker Analytics — Boston, MA
Data Analyst Intern — Jun 2026 – Aug 2026

- Architected a fully local agentic financial analyst over SEC 10-K filings, pairing a Chroma RAG layer with a separate Python calculation tool so the model never generated its own figures.
- Forecast insurance commission trends using Holt-Winters, SARIMAX, Prophet, and LSTM, benchmarking accuracy with walk-forward validation on MAPE, MAE, and RMSE; the best model reached 12% MAPE, informing client revenue planning.
- Ran year-over-year time series analysis across 18 monthly account-level files, adding reconciliation and anomaly checks that surfaced retention trends by carrier and plan type.

Hapsun Medical Solutions Pvt Ltd — Mumbai, India
Analytics Consultant — Jul 2024 – Jul 2025

- Designed and automated Tableau dashboards tracking 8 operational and revenue KPIs across 3 medical centers, eliminating 40+ hours of manual reporting per quarter and enabling data-driven growth strategies for center leadership.
- Built comparative evaluation matrices spanning 3+ options across 100+ medical equipment categories, sourcing quotes from 50+ suppliers to support procurement decisions and vendor negotiations for an incoming 1,000-bed hospital.
- Managed a 3-person team building a MongoDB-backed clinic platform, designing data models for patient, doctor, department, and appointment workflows and implementing indexed queries for patient lookup, doctor availability, and token management.

Deloitte & Touche LLP — New York City, NY
Cyber Risk Consultant — Jun 2023 – Jun 2024

- Configured data ingestion pipelines connecting 8 enterprise sources to a data discovery platform (BigID) for a large hospital system, scanning 6M+ objects to build a metadata repository covering data lineage, PII classification, and governance.
- Led stakeholder communication across the platform rollout, presenting progress to senior client executives and steering a 3-person team through technical escalations to deliver on time, earning recognition from the client as their “unsung hero.”
- Eliminated manual intake bottlenecks for 200–500 monthly privacy requests by deploying a governed self-service privacy portal, automating end-to-end processing of access, deletion, and consent workflows.
- Spearheaded delivery of a global 20+ person Quantum Risk engagement, owning trackers, budgets, timelines, and KPIs, and leading its eminence and outreach workstream, which included a client networking event for 40+ attendees on quantum risk exposure across industry sectors.

Deloitte & Touche LLP — New York City, NY
Cyber Risk Analyst — Aug 2021 – May 2023

- Built and maintained Agile compliance dashboards in Excel tracking progress across 10 security processes for nearly 400 applications migrating from on-premise servers to cloud, surfacing bottlenecks that accelerated migration by 15%.
- Reduced application technical debt by 98% ahead of cloud migration, running 400+ vulnerability scans with Checkmarx and Prisma and partnering with application teams to prioritize and remediate findings.
- Oversaw 5 workstreams, 2 of them multi-year, across 4 to 10 person client and Deloitte teams, coordinating timelines and deliverables while presenting to senior stakeholders.
- Customized a privacy risk scoring framework and evaluated 6 pilot applications through stakeholder engagement and documentation review; subsequently adopted firm-wide to standardize risk assessments across client engagements.
- Devised 25+ technical project overviews to strengthen firm ties with a major cloud provider as part of a firm initiative.

PROJECTS

Payment Protection Insurance Credit Default Model | Python | LightGBM | Scikit-learn | SHAP | Time Series Analysis

- Engineered 2,000+ behavioral time-series features from 13 months of statements for 50K AmEx customers; benchmarked Logistic Regression, Random Forest, and LightGBM, reaching 0.957 test AUC and 0.934 with the latest 6 months withheld.

Heat's Toll on Health | Python | R | Streamlit | Scikit-learn | Groq (Llama 3) | ggplot2

- Modeled 23 years of CDC environmental and hospitalization data across 31 states with distributed lag non-linear models to identify a 32% rise in heat-related hospitalizations, then deployed a Streamlit dashboard with a Groq-powered AI agent so non-technical stakeholders could query the findings directly.

SKILLS

ML & Statistics: Predictive analytics, forecasting models, time series analysis, supervised & unsupervised learning, LightGBM, SHAP, feature engineering, walk-forward validation, regression, EDA
Languages & Tools: Python, SQL, R, MongoDB, Docker, AWS, Azure, GitHub
AI & LLM Systems: RAG pipelines, agentic workflows, LLM tool-calling, Ollama, Groq, prompt engineering, n8n
BI & Visualization: Tableau, Power BI, Streamlit, Excel

TEACHING & LEADERSHIP

Columbia University New York City, NY

Teaching Assistant, Banking & Insurance Sep 2026 – Present

- Support a 50-student graduate course on banking and insurance, holding office hours and grading coursework on risk transfer, claims frequency and severity, mortgages, interest rates, time series, and catastrophe risk.

Graduate Assistant, Student Affairs, Data Science Institute Aug 2026 – Present

- Supported MSDS Orientation programming for an incoming cohort of 250 students, coordinating logistics and session delivery alongside the Student Affairs team.

Teaching Assistant, Climate Risk Assessment Jan 2026 – May 2026

- Designed a tabletop simulation of NYC climate emergencies for 40+ graduate students mapping cascading impacts across government, healthcare, utilities, and transportation; guided Python-based climate risk modeling across the full cohort.

Social Chair, DSI Student Council Jan 2026 – Present

- Plan and execute 5+ events per semester for a 300+ member graduate community, managing budgets, vendors, and logistics.

HONORS & AWARDS

2x Applause Award, Deloitte & Touche LLP 2021 – 2023

- Awarded twice for excellence in client delivery and firm initiative work.

Dean's List, Purdue University 7 of 8 semesters